DATA VORTEX — ROUND 1 (PHASE 2)

INSIGHT REPORT

Theme: Rebuilding the Social Engine

Participant details:

Team Name: Joy

Team Size: 1

Team Head: Thathi Reddy Jivan Reddy

Selected Questions: E3, M5, H4

Database: MySQL 8.0.46

Schema: data_vortex

Analytical qusetions	M5 qualifying posts	H4 qualifying users	Highest platform avg.engagement
3	20	16	3,669.38

REPORT SCOPE

This Phase 2 analysis uses SQL-based analysis of the cleaned Social Engine dataset to identify platform-level engagement differences, unusual sharing behaviour, and highly engaged users with relatively low follower counts.

The analysis covers

E3 — platform engagement comparison

M5 — unusual sharing behaviour

H4 — high-engagement users with a follower constraint.

The work demonstrates aggregation, conditional filtering,

calculated metrics, JOINs, CTEs, and window functions.

EXECUTIVE TAKEAWAY

Instagram leads average total engagement at 3,669.38.
M5 identifies 20 posts where shares exceed likes plus comments, while H4 identifies 16 users with fewer than 5,000 followers who fall within the top 10% engagement group.

Platform Engagement Insight

Objective: Compare average engagement across social media platform

Instagram average total engagement	Twitter average total engagement
3,669.38	3,563.75

Platform results

Platform	Average Total Engagement
Instagram	3,669.38
Reddit	3,647.47
YouTube	3,638.03
Facebook	3,631.00
Twitter	3,563.75

Insight

Instagram achieved the strongest average total engagement among the analysed platforms. Reddit and YouTube followed closely, while Twitter recorded the lowest average total

engagement among the listed platforms.

Analytical interpretation

The relatively small difference between the top platforms indicates that engagement levels are broadly competitive across platforms. However, Instagram shows the strongest average combined interaction in this dataset.

Unusual Sharing Behaviour

Objective: identify posts where the number of shares greater than the combined number of likes and comments.
Condition: Shares > Likes + Comments

Posts satisfying the condition	Highest total engagement in identified set
20	3,936

Highest-engagement post in the identified set

Metric	Value
Likes	981
Shares	1,997
Comments	958
Total Engagement	3,936

Insight

These posts demonstrate an unusual sharing pattern where users shared the content more frequently than the combined volume of likes and comments.

Analytical interpretation

A high share-to-interaction pattern can indicate content that users are more likely to distribute or forward rather than simply react to. These posts may represent content with stronger redistribution behaviour and can be considered potential behavioural anomalies for further investigation.

SQL analysis lens

M5 uses a calculated comparison between shares and the combined likes-plus-comments measure, followed by filtering to retain only posts meeting the stated condition.

High-Engagement Users

Objective: identify users with fever then 5,000 follower who belong to the top 10% engagement group.

Users in final result	Highest total engagement	Followers of highest-engagement users	Posts by highest-engagement users
16	64,753	1,824	16

Highest engagement users

Field	Result
User ID	user_uerv85na
Location	Rome, Italy
Followers	1,824
Posts	16
Total Engagement	64,753

The analysis joins the users and posts tables using user_id, calculates total engagement for each user, ranks users using a window function, and applies the follower-count condition.

Result profile

All selected users had follower count below 5,000 and an engagement percentile within the top 10%. Other high-performing users were distributed across locations including France, Brazil, South Africa, USA, UAE, UK, Canada, Italy, Japan, Australia, Spain, Egypt, and Canada.

Insight

The results show that high engagement is not limited to users with very large follower counts. Several users with relatively modest follower bases achieved exceptionally high total engagement.

Analytical interpretation

This suggests that follower count alone may not be a sufficient indicator of social-media performance. Engagement efficiency and content activity can help identify high-performing users who may otherwise be overlooked based only on audience size.

Cross-Question Insights

Lens	Finding
Platform performance	Instagram recorded the highest average total engagement at 3,669.38
Sharing behaviour	20 posts had shares greater than the combined likes and comments.

Audience size vs. engagement	H 4 demonstrate that users with relatively small follower count can generate substantial engagement.
User behaviour	16 users with fever then 5000 followers ranked within the top 10% engagement group.
SQL-enabled analysis	The three questions collectively cover platform-level ,post-level and user-level analysis

Key findings

• Instagram had the highest average total engagement: 3,669.38.

• 20 posts showed shares greater than likes plus comments.

• The highest-engagement M5 post recorded 3,936 total engagement.

• 16 users with fewer than 5,000 followers appeared in the top 10% engagement group.

• The highest-engagement H4 user generated 64,753 total engagement from 16 posts.

• High engagement can occur even among users with relatively modest follower counts.

Analytical coverage

E3 - platform-level

M5 - post-level

H4 - user-level

Conclusion

The Phase 2 SQL analysis successfully converts the cleaned Social Engine dataset into actionable insights at platform, post, and user

levels.

Qusetions	Analytical contribution
E3	Highlights the strongest platform-level engagement
M5	Identifies unusual sharing behaviour that may represent distinctive patterns
H4	Demonstrates that strong engagement can be achieved by users without extremely large followers bases

Together, these findings show how SQL based analytical techniques can reveal meaningful patterns in social media data and support d for behaviour analysis

Methods Used

Technique	Role in analysis
Aggregation	Summary engagement across analytical levels.
Conditional filtering	Isolated post/users meeting defined conditions.
Calculated metrics	Derive total engagement and comparison measures.
JOINs	Connect users and post throw user_id.
CTEs	Texture multi-step SQL analysis.
Window function	Rank uses for percentile-based selection.